

Problem Set 3

This abbreviated pset is due at 11am on **Friday Sep 30**.

Problem 1 (5+5 points)

(a) Give the rank of D and a basis for the nullspace $N(D)$ for the D matrix from pset 1:

$$D = \begin{pmatrix} -\frac{1}{\Delta x} & \frac{1}{\Delta x} & 0 & 0 & 0 \\ 0 & -\frac{1}{\Delta x} & \frac{1}{\Delta x} & 0 & 0 \\ 0 & 0 & -\frac{1}{\Delta x} & \frac{1}{\Delta x} & 0 \\ 0 & 0 & 0 & -\frac{1}{\Delta x} & \frac{1}{\Delta x} \end{pmatrix}$$

(Does your answer depend on the value of Δx ?)

(b) Consider the vector space V consisting of all differentiable functions $f(x)$, and let $A = \frac{d}{dx}$ be the linear operator that takes the derivative, i.e. $Af = f'$ for any $f \in V$. Give a basis for $N(A)$, the nullspace of the derivative. Is there a resemblance to your answer in part (a)?

Problem 2 (5+5 points)

In class, we considered elimination on the matrix $A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 5 & -3 \\ 1 & 2 & 7 & -7 \end{pmatrix}$, and found

that it was rank 2 with the pivot and free columns "interleaved" (the first and third columns were pivot columns).

(a) Give a permutation matrix P such that doing Gaussian elimination on AP (which re-orders the _____ of A) leads to the first two columns being the pivot columns (i.e. "non-interleaved" pivot and free columns).

(b) How do the nullspace $N(AP)$ and column space $C(AP)$ relate to the null and column spaces of A , respectively? In class, a basis for $N(A)$ was the two special solutions $[-2, 1, 0, 0]$ and $[-7, 0, 2, 1]$ and from these how can you get a basis for $N(AP)$? (Note: the problem gave the nullspace vector as $[-7, 0, 3, 1]$, following a typo from the notes. You won't be penalized if you copy this and use 3 instead of 2 here.)

Problem 3 (5+5 points)

In class, we saw that *invertible* row operations (such as those occurring in Gaussian elimination) do *not* change the null space of a matrix.

(a) What happens if you do *non*-invertible row operations $B = EA$? Does that generally lead to $N(B) = N(A)$, or $N(B) \subseteq N(A)$ (a *smaller* nullspace), or $N(B) \supseteq N(A)$ (a *larger* nullspace), or none of these (neither nullspace contains the other)? Justify your answer.

(b) Illustrate your answer in (a) with example A , E , and B matrices, and give bases for the nullspaces of your A and B .

Problem 4 (3+3+3+3 points)

(from Strang, section 3.3)

Give examples of matrices A for which the number of solutions to $Ax = b$ is

(a) 0 or 1, depending on b .

(b) ∞ , regardless of b .

(c) 0 or ∞ , depending on b

(d) 1, regardless of b .